



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE423: Simulation and Modeling

Batch: 23rd

Semester: Spring 2026

Date: 10-06-2026

Mid-term Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

(স্বাক্ষরিত মূল্য) $5 \times 6 = 30$

Answer any five questions from the followings.

1. (a) Define any three terminologies of system simulation with an example. 3
(b) Draw the flowchart for the steps involved in simulation study. 3
2. (a) A coffee shop wants to simulate customer service. Which elements must be included in the discrete-event simulation model? 3
(b) What are the common problems faced during verification and validation? 3
3. (a) How is the time advance mechanism used in discrete-event simulation? 3
(b) A hospital wants to model each patient's journey from registration to doctor consultation and discharge. Which approach is better; event scheduling or process interaction? 3
4. (a) How is probability involved in discrete event simulation? 2
(b) Illustrate the four activities that occur inside the single-server queueing system. 4
5. (a) Briefly explain different types of perspectives for modeling of a system. 3
(b) Construct a diagram for arrival routine in queueing system. 3
6. (a) In an M|M|1 queue if $\lambda = 9$ and $M = 12$, find the W and W_q . 2
(b) In time series validation:
Real: [3, 4, 8, 10, 16]
Simulation: [2, 3, 8, 9, 14]
Calculate mean Absolute Percentage Error (MAPE) and check the validity. 4
7. In an M/M/1/K queueing system with a single server, arrival rate $\lambda = 3/\text{hr}$, service rate $\mu = 4/\text{hr}$, system capacity $k=4$. Find the steady possibilities $P(n)$, Blocking possibility P_k , Average number in system L and effective arrival rate λ_{eff} and loss Rate. 6



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 421: Software Engineering 307441

Batch: 23rd
Date: 08/06/2026

Mid-Term Examination

Semester: Spring 2026
Time: 1 Hour 30 Minutes
Total Marks : 30

অনোয়াড় সাজা

5 x 6 = 30

Answer any **five** questions:

1. (a) Describe the unique nature of Web Applications.
(b) Explain the general principles of software engineering. ✓
2. (a) Illustrate different types of process flow. ✓
(b) Describe three types of process pattern. ✓
3. (a) Explain the software process framework activities.
(b) Create a task set for a large and complex software project. ✓
4. (a) Explain the human factors in agile process.
(b) What is Agility? List all Agility principles.
5. (a) Draw a table for Extreme programming practices.
(b) Describe Industrial Extreme Programming (IXP).
6. (a) Create a table comparing different agile techniques. ✓
(b) Briefly explain the stages involved in DevOps approach.
7. Write short notes on the following:
I. Product Backlog ①
II. Burn Down Chart ②
III. Scrum Master ③

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STANDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
MGT 417: Industrial and Operational Management

Batch: 23rd

296389

Mid Term Examination

Semester: Spring 2026

Date: 20/06/2026

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

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5 x 6 = 30

1. (a) Define Industrial Management. 1
(b) Briefly explain the levels of an Industry. 5
2. (a) Write the importance of Management. 3
(b) Explain the functions of Management with diagram. 3
3. (a) State the needs, scope and applications of Industrial Management. 3
(b) Briefly explain the hierarchy of different levels of Manager. 3
4. (a) State and explain the managerial skills. 2
(b) Explain the managers at different areas of an organization. 4
5. (a) State the role of an Office Manager. 2
(b) "Management is an arts or science", Justify. 4
6. Briefly explain the terms production, productivity, productivity index and factors affecting the productivity of an organization. 6
7. A furniture manufacturing company has provided the following data. Compare the labor, raw materials and supplies, capital equipment depreciation, other and total productivity of 1996 and 1997. 6
Output: Sales value of production in dollar (\$) 22,000 (in 1996) and 35,000 (in 1997).

Inputs	1996	1997
Labor	10,000	15,000
Raw materials and supplies	8,000	12,500
Capital equipment depreciation	700	1,200
Other	2,200	4,800



হামদার স্টাড HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE 469 : Digital Image Processing

Batch: 23rd

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Semester: Spring 2026

Date: 17/06/2026

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions.

$5 \times 6 = 30$

- (a) Define the following terms: 2

~~(i) Pixel~~ (i) Digital Image

(b) Briefly describe the fundamental steps in Digital Image Processing. 4
- ~~(a) Define Sampling and Quantization.~~ (a) Define Sampling and Quantization. 2

(b) Explain the Simple Image Formation Model. 4
- (a) Write short notes on - 3

~~(i) Neighbors of a Pixel and~~ (i) Neighbors of a Pixel and

~~(ii) Adjacency of a Pixel.~~ (ii) Adjacency of a Pixel.

(b) What is the ambiguity problem associated with 8-adjacency and how can it be resolved? Explain using an example. 3
- (a) What is the Histogram of an Image? Mention the objectives or goal of Histogram Equalization. 2

(b) Suppose that a 2-bit image ($L = 4$) of size 32×32 pixels ($MN = 1024$) has the following intensity distribution, where the intensity levels are integers in the range $[0, L - 1] = [0, 3]$. 4

Intensity r_k	Pixel Count n_k
$r_0 = 0$	374
$r_1 = 1$	102
$r_2 = 2$	126
$r_3 = 3$	422

Now, compute the equalized histogram values using the histogram equalization transformation function and illustrate the original histogram and equalized histogram.

- (a) Use the following 3×3 mask to perform the convolution process on the shaded pixels in the 5×5 image below. Write the filtered image pixel values. 3

0	1/4	0
1/4	1/2	1/4
0	1/4	0

3×3 mask

30	40	50	70	90
40	50	80	60	100
35	255	70	0	120
30	45	80	100	130
40	50	90	125	140

5×5 image

(b) Given the following 1-D image strip:

5	5	4	3	2	1	0	0	5	0	0	0	1	3	1	0	0	0	7	7	7
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3

Compute the first and second derivatives and using your results, briefly explain how the first and second derivatives respond to thick edges, thin edges and gray-level steps.

6. (a) Explain why do we need Fourier transform in digital image processing. 2
- (b) Distinguish between Spatial and Frequency domain filtering techniques. 2
- (c) Mention the steps involved in filtering in the frequency domain. 2
7. (a) Explain different types of Smoothing (Lowpass) frequency domain filters. 3
- (b) Differentiate between Smoothing and Sharpening filtering techniques. 3